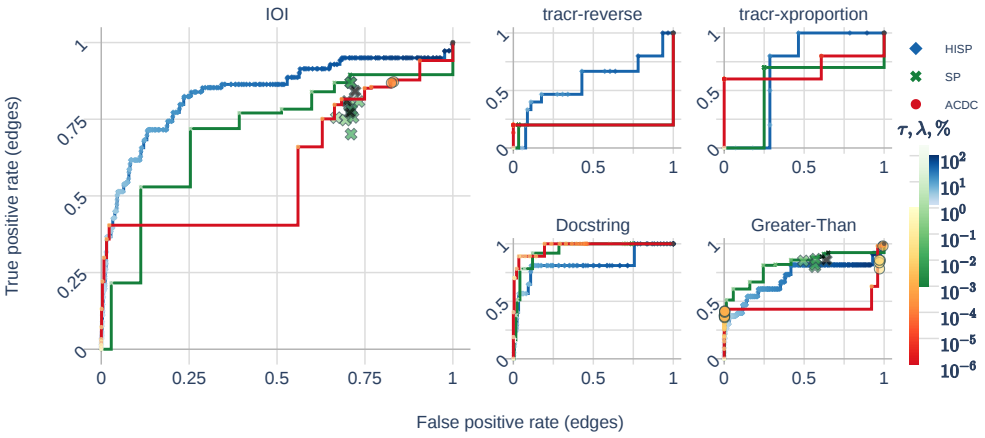

Figure 7: Edge-wise ROC curves generated by minimizing the task-specific metric in Table 1, rather than KL divergence.

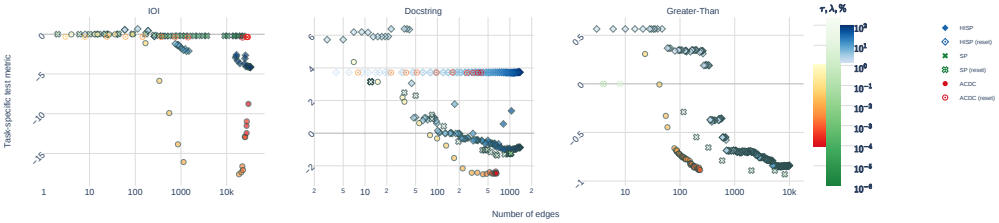

Figure 8: Optimizing the task-specific metric of the subject model, on trained and reset networks. For each recovered circuit, we plot its task-specific metric (Table 1) against its number of edges. The reset networks metrics don't change much with the number of edges, which is good. We found that for IOI that extremely large logit differences could be achieved (over 15) but this didn't happen when the network had a large number of edges.

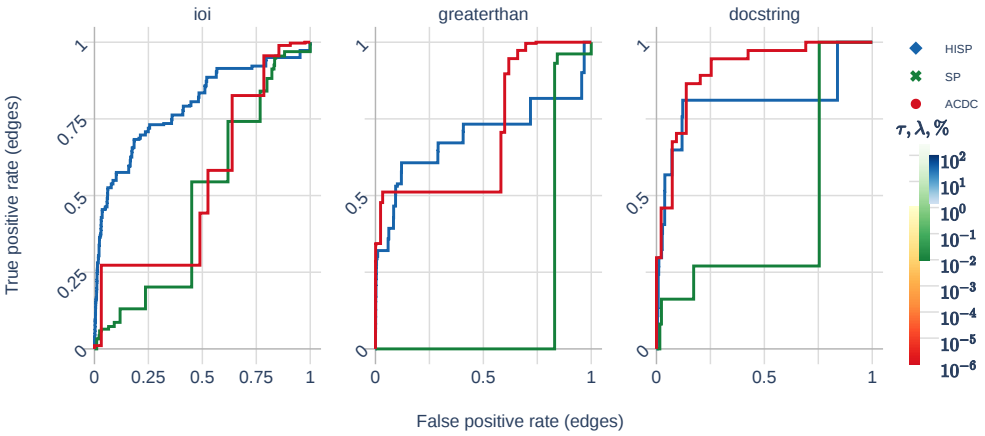

Figure 9: Edge-wise ROC curves generated by minimizing the KL divergence, but using zero activations.

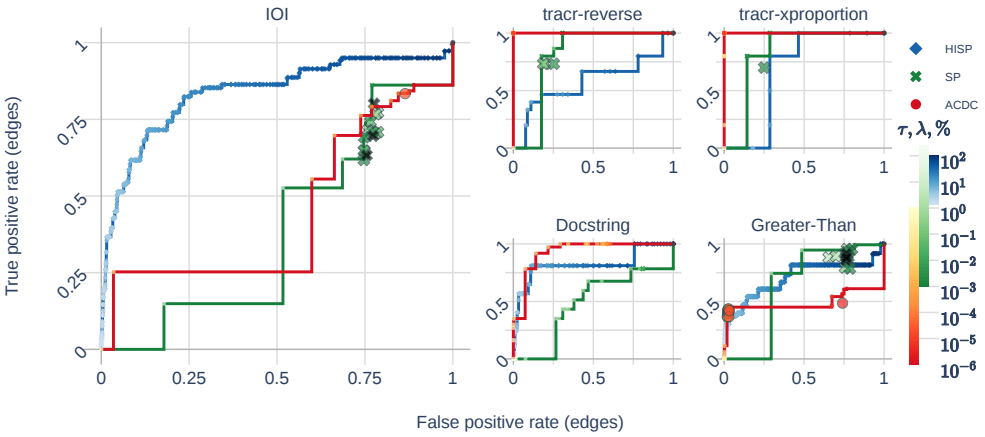

Figure 10: Edge-wise ROC curves generated by minimizing the task-specific metric in Table 1, using zero activations.

E.3 Node-level ROC curve, rather than edge-level ROC curve

We compute the FPR and TPR of classifying whether a node belongs to the circuit. We consider this alternative task because SP and HISP operate at the node-level, whereas ACDC operates at the edge-level, so this is fairer to HISP and SP. The results are broadly similar to edge-level ROCs, and are described in Figs. 11 and 12 and Tables 2 and 3.

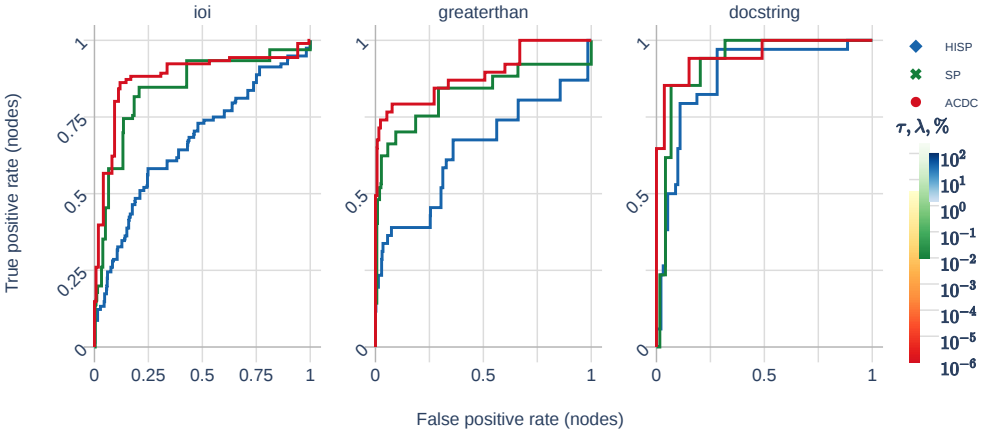

Figure 11: Node-level ROC when optimizing the KL divergence, with corrupted activations.

F IOI task: details and qualitative evidence

F.1 Further details on the IOI experiments

In the ACDC run in Figure 1, we used a threshold of $\tau = 0.0575$. We also removed all edges which did not lie on a directed path from the input (which is equivalent in computation since we use corrupted activations). Our library now only supports splitting query, key and input, rather than merely looking at the connections between heads. Additionally, For ease of visualization, in the diagram on the left of Figure 1 we removed all edges between grey nodes more than 2 layers apart, and 90% of the edges between grey and red nodes.

Our IOI experiments were conducted with a dataset of $N = 50$ text examples from one template the IOI paper used (‘ When John and Mary went to the store, Mary gave a bottle of milk to’). The

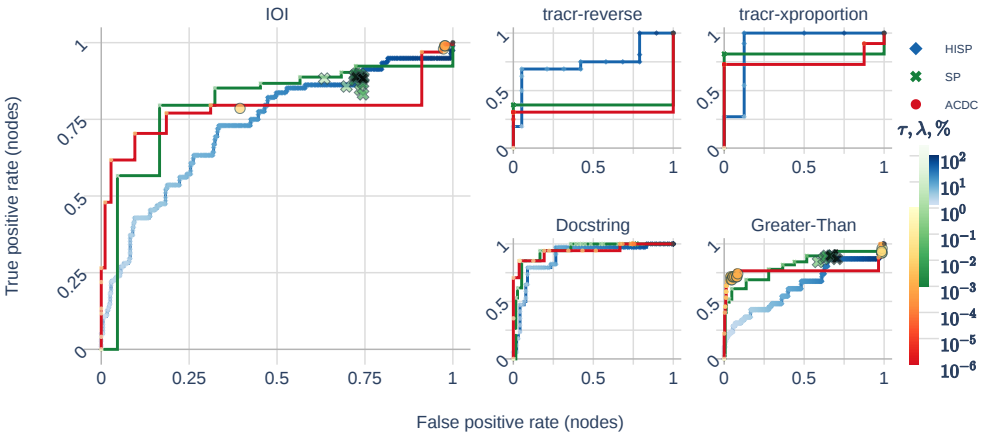

Figure 12: Node-level ROC when optimizing the task-specific metric in Table 1, with corrupted activations.

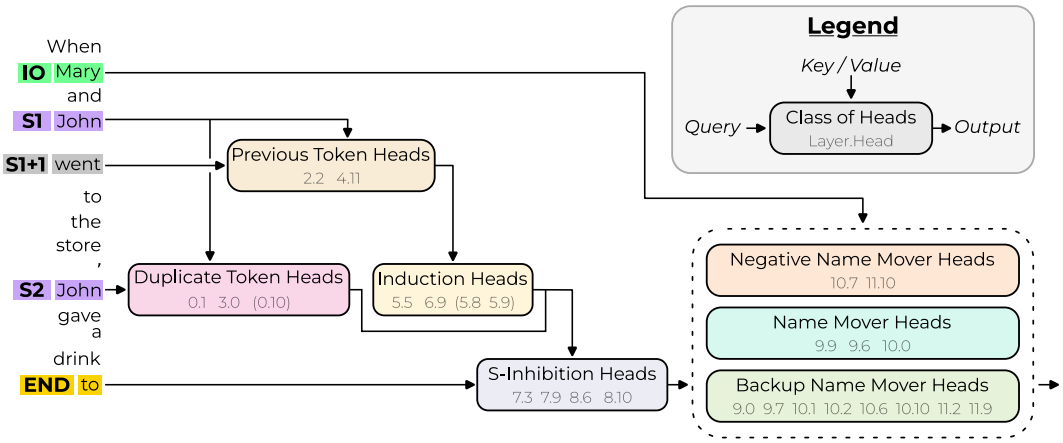

Figure 13: The IOI circuit (Figure 2 from Wang et al. (2023)). We include edges between all pairs of heads connected here, through the Q or the K+V as indicated. We also include all connections between MLPs and these heads, the inputs and outputs. See `https://github.com/ArthurConmy/Automatic-Circuit-Discovery/blob/main/acdc/ioi/utils.py#L205`. The full circuit is in Fig. 14.

corrupted dataset was examples from the ABC dataset (Wang et al., 2023) — for example 'When Alice and Bob went to the store, Charlie gave a bottle of milk to'.

In the IOI experiment in Figure 1, we did not split the computational graph into the query, key and value calculations for each head. This enabled the ACDC run to complete in 8 minutes on an NVIDIA A100 GPU. However, the larger experiments that kept >10% of the edges of the original edges in the computational graph sometimes took several hours. On one hand we don't expect these cases to be very important for circuit discovery, but they make up the majority of the points of the pareto frontier of curves in this paper.

F.2 The IOI circuit

Wang et al. (2023) find a circuit 'in the wild' in GPT-2 small (Radford et al., 2019). The circuit identifies indirect objects (see for example Table 1) by using several classes of attention heads. In this subsection we analyze how successful ACDC's circuit recovery (Figure 1) is. All nine heads found in Figure 1 belong to the IOI circuit, which is a subset of 26 heads out of a total of 144 heads in GPT-2 small. Additionally, these 9 heads include heads from three different classes (Previous Token Heads,